

Systematic Sampling: 1999 Q6(a), 2001

Systematic sampling is operationally more convenient than simple random sampling. This selection procedure is different from simple random sampling procedure in the context that in simple random sampling procedure every unit is selected with the help of random numbers whereas in the systematic ~~sampling~~ selection only the first unit is to be selected at random and the remaining units are automatically determined by the skip interval.

Suppose that there are N units in the population numbered from 1 to N , if $N/n = k$, where " n " denotes the number of units in the sample and k (skip interval) is an integer.

The population of N units would be divided into ' n ' groups each containing ' k ' units.

Systematic sampling is a technique to select a sample from system.

It is called systematic sampling because "it is sampling is done on the basis of pre-determine system."

In systematic sampling only first unit

Only start is random but thereafter we select sample by pre-determined system

is selected randomly. while the selection of all other units depends on the selection of first unit, so systematic sampling is not truly randomized sampling, so, it can not be called systematic random sampling.

Systematic Sampling Procedure Requires:

- 1) Number all the sampling unit in the population as $1, 2, \dots, N$
- 2) Divide the population into a number of "n" groups each of size k
- 3) Decide about the size of the group (k) where $k = \frac{N}{n}$
- 4) Draw a random number let 1-k and every k^{th} unit thereafter will be selected for the sample.

→ Linear Systematic Sampling. 1996 Q6(a)+2001
e.g. $N=20$, $k=5$, $n=4$

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20

I					II					III		
1	2	3	4	5	6	7	8	9	10	11	12	13
14	15	16	17	18	19	20						

All the sampling units in interval I has equal probability of being selected.

Suppose a random number selected from interval I is r then the systematic sample will be

S.No. Systematic Samples

1. 2, 7, 12, 17 (Suppose no selected is 2)

2. 3, 8, 13, 18 (" " " " 3)

3. 1, 6, 11, 16 (" " " " 1)

4. 5, 10, 15, 20 (" " " " 5)

5. 4, 9, 14, 19 (" " " " 4)

⇒ This implies that the selection of first sampling unit determines the selection of whole sampling units to be included in the sample.

⇒ In systematic sampling all possible samples is equal to k (length)

i.e;

$$A.P.S = k$$

As in our example $k=5$, So that no. of systematic samples are also 5.

This procedure is called Linearly Systematic Sampling. In this type of systematic sampling N is an integral multiple of nk when $N=nk$ and each sample has equal size.

When $N = nk$ (mean N is integral multiple of nk) then each sample is of same equal size

→ Circular ~~Systematic~~ Sampling: ^{systematic}
When $N = nk$, then linear sampling is used

But when N is not integral multiple of nk i.e; $N \neq nk$. Then we will go for circular systematic sampling.

e.g: $N = 23$, $n = 5$, $k = \frac{23}{5} = 4.6 \approx 5$

I	II	III	IV	V
1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	(1)	(2)

Sample size of I, II, III = 5 while sample size of IV and V = 4. Means sample size does not remain same while it varies because $N \neq nk$.

And;

→ This difference is always 1, it will never be greater than 1.

⇒ When $n \geq 50$ then;

1. we ignore the difference, and the large

number of sample size (which sample size comes more) is taken as fixed size. but when n is small then $\rightarrow$

OR

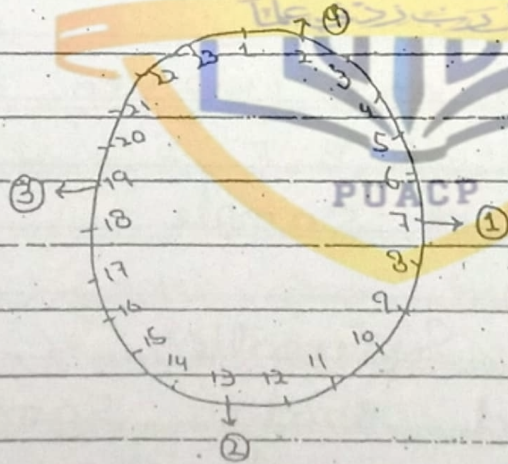
2) We complete the sample (means equalize each sample size) by repeating some first values and still sample remains different from other means each and every sampling unit must appear once and only once

$\rightarrow$ Problem of unequal sizes can be solved by using circular systematic sampling which is suggested by Lahiri.

e.g.:-

Draw a random number b/w 1, S , N

Suppose the
s.no. selected
is 7



$$n=4, k=5:5=6$$

Thus systematic sample is 7, 13, 19, 5
 $1999 \quad 26(4) + 2001$

$\Rightarrow$ Centrally Located Systematic Sampling

When sample is selected from centre of group then 1st sampling unit lies in the centre of 1st group then all other sampling unit will lie in the centre of their own group.

⇒ This implies that location of first central sampling unit determines the location of whole central sampling units to be included in the sample.

This type of sample is called centrally located sample and the procedure is called Centrally located Systematic Sampling.

When $k = \text{odd}$:

Then $(\frac{k+1}{2})^{\text{th}}$ sampling unit will be —

the central sampling unit.

When $k = \text{even}$:

Then $(\frac{k}{2})^{\text{th}}$ and $(\frac{k+2}{2})^{\text{th}}$ both are

centrally located sample.

✓ Comparison Of Systematic & Stratification.

Stratified Random Sampling is more precise than Systematic Sampling while Systematic Sampling is more precise than simple Random Sampling.

Also systematic Sampling is convenient, less time consuming, easy and cheap.

While it is biased in some situation e.g. If in a factory Quality inspection team came then sample is selected systematically by choosing first unit randomly. But in such case factory owner wants to prevent such sampling units.

ISO: International Standard Organization

which are quality-wise better than others.
So to select best units, which results in
biasedness

Examples: When k is odd

Suppose when $N=20$, $n=4$, $k=5$

① centre
1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16
17, 18, 19, 20
④ centre
③ centre
 $\frac{20}{2} = 10$

Central

All possible

sample

i.e., 3, 8, 13, 18

When k is even

$N=16$

$n=4$

$k=4$

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1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16

When k is even then there will be 2 central
sampling units from which one can be chosen

i.e.

We can choose either $\left(\frac{k+2}{2}\right)^{th} = \left(\frac{4+2}{2}\right)^{th} = 3^{rd}$ unit
will be central unit and the sample will
be: 3, 6, 9, 12

or $\left(\frac{k}{2}\right)^{th} = \left(\frac{4}{2}\right)^{th} = 2^{nd}$ unit will be the central

sampling unit then sample will be 2, 5, 8, 11

Notations: Σ Lay-out:

y_{ij} - The observation for j th sampling unit in the i th systematic sample

$$N = nk$$

$$i = 1, 2, 3, \dots, k$$

$$j = 1, 2, 3, \dots, n$$

Systematic Sample No

N	I	II	...	i	...	k	St. mean $\bar{y}_i$
1	$(y_{11} - \bar{y}_{\cdot 1})$ y_{11}						$\bar{y}_{\cdot 1}$
2	$(y_{21} - \bar{y}_{\cdot 2})$ y_{21}						$\bar{y}_{\cdot 2}$
3	y_{31}						
...	...						
i	y_{ij}			$(y_{ij} - \bar{y}_{\cdot i})$ y_{ij}			$\bar{y}_{\cdot i}$
...	...						
n	y_{in}						
$\sum_j y_{ij}$	$y_{i\cdot}$	$y_{2\cdot}$		$y_{i\cdot}$		$y_{k\cdot}$	
$\bar{y}_{i\cdot}$	$\bar{y}_{\cdot 1}$	$\bar{y}_{\cdot 2}$		$\bar{y}_{\cdot i}$		$\bar{y}_{\cdot k}$	

$$y_{n\cdot}$$

$$y_{i\cdot} = \sum_j y_{ij} = \text{Total for } i\text{th systematic sample}$$

$$\bar{y}_{i\cdot} = \frac{y_{i\cdot}}{n} = \frac{\sum_j y_{ij}}{n} = \text{The mean of } i\text{th systematic sample}$$

$$\bar{y}_{sys} = \frac{1}{k} \sum \bar{y}_{i\cdot} = \text{The estimated mean for systematic sampling}$$

$$S_{wsys}^2 = \frac{1}{k(n-1)} \sum_i \sum_j (y_{ij} - \bar{y}_{i\cdot})^2 = \text{Variance within systematic samples}$$

Theorem:

Show that the variance of the mean of the systematic sample is

$$V(\bar{y}_{sy}) = \frac{(N-1)S^2}{N} - \frac{k(n-1)S_{wsy}^2}{N}$$

Proof:

By the identity of analysis of variance

As we know that

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y})^2$$

(never write sum of n before summation of k)

Adding & Subtracting $\bar{y}_{i.}$

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.} + \bar{y}_{i.} - \bar{y})^2$$

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n [(y_{ij} - \bar{y}_{i.}) + (\bar{y}_{i.} - \bar{y})]^2$$

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n [(y_{ij} - \bar{y}_{i.})^2 + (\bar{y}_{i.} - \bar{y})^2 + 2(\bar{y}_{i.} - \bar{y})(y_{ij} - \bar{y}_{i.})]$$

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 + \sum_{i=1}^k n(\bar{y}_{i.} - \bar{y})^2 + 2(\bar{y}_{i.} - \bar{y}) \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})$$

The cross product term reduces to zero as the different systematic samples are taken independently

$$(N-1)S^2 = \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 + \sum_{i=1}^k n(\bar{y}_{i.} - \bar{y})^2 \quad \text{--- (I)}$$

$$\Rightarrow S_{wsy}^2 = \frac{1}{k(n-1)} \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

$$\Rightarrow S_{wsy}^2 = \frac{1}{n(k-1)} \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

We know that

$$S_{wsy}^2 = \frac{1}{k(n-1)} \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2$$

$$\Rightarrow \sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{i.})^2 = k(n-1) S_{wsy}^2$$

Put this result in (I) we get

$$(N-1)S^2 = k(n-1) S_{wsy}^2 + n \sum_{i=1}^k (\bar{y}_{i.} - \bar{y})^2 \quad \text{--- (II)}$$

We define:

$$V(\bar{y}_{sy}) = \frac{\sum_{i=1}^k (\bar{y}_{i.} - \bar{y})^2}{k} \quad \text{(By definition)}$$

$$\Rightarrow \sum_{i=1}^k (\bar{y}_{i.} - \bar{y})^2 = k V(\bar{y}_{sy})$$

Put this value in (II), we get

$$(N-1)S^2 = k(n-1) S_{wsy}^2 + nk V(\bar{y}_{sy})$$

$$\text{As } N = nk$$

$$(N-1)S^2 = k(n-1) S_{wsy}^2 + N V(\bar{y}_{sy})$$

Dividing both sides with 'N'

$$\left(\frac{N-1}{N} \right) S^2 = \frac{k(n-1)}{N} S_{wsy}^2 + V(\bar{y}_{sy})$$

$$V(\bar{y}_{sy}) = \left(\frac{N-1}{N} \right) S^2 - \frac{k(n-1)}{N} S_{wsy}^2 \quad (\text{proved})$$

✓ Theorem: 1999 Q6(a) + 2007

Show that the mean of a systematic sample is more precise than a mean of a simple random sample if and only if $S_{wsy}^2 > S^2$

Proof: The mean of a systematic sample will be more precise than a mean of a simple random sample if $V_{sy} < V_{ran}$ $\rightarrow V_{sy} < V_{ran}$

$$V(\bar{y}_{sy}) = \left(\frac{N-1}{N} \right) S^2 - \frac{k(n-1)}{N} S_{wsy}^2$$

And

$$V_{ran} = \left(\frac{N-n}{N \cdot n} \right) S^2$$

$$\Rightarrow \left(\frac{N-1}{N} \right) S^2 - \frac{k(n-1)}{N} S_{wsy}^2 < \frac{N-n}{N \cdot n} S^2$$

$$-k(n-1) S_{wsy}^2 < \left(\frac{N-n}{n} \right) S^2 - (N-1) S^2$$

$$-k(n-1) S_{wsy}^2 < \left[\frac{N-n-Nn+n}{n} \right] S^2$$

Put $N = nk$ in R.H.S

$$-k(n-1) S_{wsy}^2 < \left(\frac{nk - n^2k}{n} \right) S^2$$

stratified is more precise than SRS if variance within stratum is small and opposite is the case with systematic sampling.

$$-K(n-1)S_{w\text{sy}}^2 < (K-nK)S^2$$

$$-K(n-1)S_{w\text{sy}}^2 < -K(n-1)S^2$$

Multiply inequality with -1

$$K(n-1)S_{w\text{sy}}^2 > K(n-1)S^2$$

$$S_{w\text{sy}}^2 > S^2$$

This result shows that systematic sampling is more precise than simple random sampling if the variance within the systematic samples are is larger than the population variance as a whole.

Systematic Sampling is precise when units within the same sample are heterogeneous and is imprecise (not precise) when they are homogeneous. In fact, if there is little variation within a systematic sample relative to that in the population, the successive unit in the sample are repeating more or less (or the same) the same information.
 e.g. 1, 1, 2, 2, 2, 3, 3 then variance will be small.

Theorem: 2001 Q.6(a)

Show that vari

$$V(\bar{y}_{\text{sy}}) = \frac{S^2}{n} \left(\frac{N-1}{N} \right) [1 + (n-1)l_w]$$

where;

Theorem:- Develop an expression in terms of S^2 and ρ_w where ρ_w is the correlation coefficient b/w pairs of units in the same sample.

ρ_w is the population correlation coefficient between pairs of units that are in the same systematic sample.

Proof: By definition

$$V(y_{sy}) = \frac{1}{k} \sum_i^k (\bar{y}_{i.} - \bar{y})^2$$

$$= \frac{1}{k} \sum_i^k \left[\frac{\sum_j y_{ij}}{n} - \bar{y} \right]^2$$

$$= \frac{1}{k} \sum_i^k \left[\frac{\sum_j y_{ij}}{n} - \frac{n\bar{y}}{n} \right]^2 \quad (\times S_2 \text{ with } n)$$

$$= \frac{1}{k} \sum_i^k \left[\frac{1}{n} \sum_j (y_{ij} - \bar{y}) \right]^2 \quad \because n\bar{y} = \sum_j \bar{y}$$

$$= \frac{1}{n^2 k} \sum_i^k \left[\sum_j (y_{ij} - \bar{y}) \right]^2$$

$\therefore$ Sq. on summation open like that

$$= \frac{1}{n^2 k} \sum_i^k \left[\sum_j (y_{ij} - \bar{y})^2 + \sum_{j \neq u} (y_{ij} - \bar{y})(y_{iu} - \bar{y}) \right]$$

$\therefore$ convert $\sum_{j \neq u} \rightarrow \sum_j \sum_{j \neq u}$

$$V(y_{sy}) = \frac{1}{n^2 k} \left[\sum_i^k \sum_j (y_{ij} - \bar{y})^2 + \sum_i^k \sum_{j \neq u} (y_{ij} - \bar{y})(y_{iu} - \bar{y}) \right] \quad \text{--- (I)}$$

We define:

$$S^2 = \frac{1}{N-1} \sum_i^k \sum_j (y_{ij} - \bar{y})^2$$

$$\Rightarrow \sum_i^k \sum_j (y_{ij} - \bar{y})^2 = (N-1) S^2 \quad \text{--- (II)}$$

We define

$$e_{uw} = \frac{E(y_{ij} - \bar{y})(y_{iu} - \bar{y})}{E(y_{ij} - \bar{y})^2} \quad \therefore \text{Expectation Sum of values} \\ \text{An no. of values}$$

$$E(y_{ij} - \bar{y})^2 = \frac{\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y})^2}{nk} \quad \therefore \text{These are } n(n-1) \text{ pairs} \\ \text{of units in one systematic sample}$$

And:

$$E(y_{ij} - \bar{y})(y_{iu} - \bar{y}) = \frac{\sum_{i=1}^k \sum_{j \neq u}^n (y_{ij} - \bar{y})(y_{iu} - \bar{y})}{k n(n-1)} \quad \text{So for } k \text{ systematic samples these are} \\ \text{pairs of units}$$

Put these values in e_{uw}

$$e_{uw} = \frac{\sum_{i=1}^k \sum_{j \neq u}^n (y_{ij} - \bar{y})(y_{iu} - \bar{y}) / k n(n-1)}{\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y})^2 / nk}$$

$$e_{uw} = \frac{\sum_{i=1}^k \sum_{j \neq u}^n (y_{ij} - \bar{y})(y_{iu} - \bar{y}) (n-1)}{(N-1) S^2}$$

$$e_{uw} = \frac{\sum_{i=1}^k \sum_{j \neq u}^n (y_{ij} - \bar{y})(y_{iu} - \bar{y})}{(n-1)(N-1) S^2}$$

$$\sum_{i=1}^k \sum_{j \neq u}^n (y_{ij} - \bar{y})(y_{iu} - \bar{y}) = (n-1)(N-1) S^2 e_{uw} \quad \text{--- III}$$

Now putting results of (II) & (III) in eq (I), we get;

$$V(\bar{y}_{sy}) = \frac{1}{n^2 k} [(N-1)S^2 + (n-1)(N-1)S^2 \rho_w]$$

$$V(\bar{y}_{sy}) = \frac{1}{Nn} [(N-1)S^2 + (n-1)(N-1)S^2 \rho_w] \quad \because N=nk$$

$$V(\bar{y}_{sy}) = \left(\frac{N-1}{N \cdot n} \right) S^2 [1 + (n-1)\rho_w]$$

(proved)

~~Final Term~~

This result shows that the correlation b/w units in the same sample inflates the variance of sample mean. Even a small value of correlation may have a larger effect because of the multiply of $(n-1)$.

Theorem: 2003 Q7(b)

★ ★ ★ Show that

$$V_{\bar{y}_{sy}} = \left(\frac{N-n}{N \cdot n} \right) S_{wst}^2 \cancel{\left(\frac{N-n}{N \cdot n} \right)} [1 + (n-1)\rho_{wst}]$$

where

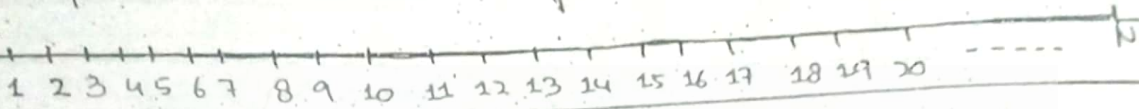
ρ_{wst} is the correlation coefficient b/w the pairs of units that are in the same systematic sample

And

$$S_{wst}^2 = \frac{1}{n(k-1)} \sum_{j=1}^{kn} (y_{ij} - \bar{y}_{.j})^2$$

Proof: This theorem is actually the relationship between stratified random sampling and systematic sampling.

By stratified random sampling is more precise than systematic sampling b/c



let say $k=10$ then first sampling unit will be selected b/w $1-k$ i.e. $1-10$

let say first S.U. is 3 then systematic sample will be:

3, 13, 23, ----

While in stratified random sampling, sampling unit is selected randomly then combining all randomly selected sampling unit to form stratified random sample.

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The drawback of systematic sampling is that if the first sampling unit is too low then other sampling units are also too low or if first S.U. is too high then other S.U. are also too high.

While in stratified random sampling sampling units are selected random so in stratified random sample there may be all units i.e. too high, too low and medium. In this type of sample, errors are balanced out. This is the good sample.

$E(y_{ij} - \bar{y}_{.j})$ = Deviation of y_{ij} from respective mean

Proof:

By definition

$$\bar{y}_{.j} = \frac{1}{k} \sum_i y_{ij}$$

$$\bar{y}_{.i} = \frac{1}{n} \sum_j y_{ij}$$

$$V(\bar{y}_{sy}) = \frac{1}{k} \sum_i (\bar{y}_{.i} - \bar{y})^2$$

Imp step

First we know that $\bar{y} = \frac{1}{n} \sum_j \bar{y}_{.j}$ $\therefore \bar{y} = \frac{1}{n} \sum_j \bar{y}_{.j}$

Now we use $\bar{y}_{.j}$ formula
If we convert $\bar{y}$

By formula $\bar{y}$ $= \frac{1}{k} \sum_i \left[\frac{1}{n} \sum_j (y_{ij} - \bar{y}_{.j}) \right]^2$

$$V(\bar{y}_{sy}) = \frac{1}{n^2 k} \sum_i \left[\sum_j (y_{ij} - \bar{y}_{.j}) \right]^2$$

$$V(\bar{y}_{sy}) = \frac{1}{n^2 k} \left[\sum_j (y_{ij} - \bar{y}_{.j})^2 + \sum_{j \neq u} (y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u}) \right]$$

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$\therefore N = nk$

$$V(\bar{y}_{sy}) = \frac{1}{N \cdot n} \left[\sum_j \sum_i (y_{ij} - \bar{y}_{.j})^2 + \sum_{j \neq u} \sum_i (y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u}) \right] \quad (I)$$

As:

$$S^2_{wst} = \frac{1}{n(k-1)} \sum_j \sum_i (y_{ij} - \bar{y}_{.j})^2$$

$$\Rightarrow \sum_j \sum_i (y_{ij} - \bar{y}_{.j})^2 = n(k-1) S^2_{wst} \quad (II)$$

We define

$$\rho_{wst} = \frac{E(y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u})}{E(y_{ij} - \bar{y}_{.j})^2}$$

$$\rho_{wst} = \frac{\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u}) / nk(n-1)}{SS(y_{ij} - \bar{y}_{.j})^2 / nk}$$

$$\rho_{wst} = \frac{\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u})}{(n-1)(k-1) S_{wst}^2} \quad \text{using II}$$

$$\sum_{i=1}^k \sum_{j=1}^n (y_{ij} - \bar{y}_{.j})(y_{iu} - \bar{y}_{.u}) = n(n-1)(k-1) S_{wst}^2 \times \rho_{wst} \quad \text{--- (III)}$$

Putting results of (II) & (III) in (I)

$$V_{ysy} = \frac{1}{N \cdot n} \left[n(k-1) S_{wst}^2 + n(n-1)(k-1) S_{wst}^2 \times \rho_{wst} \right]$$

$$V_{ysy} = \frac{n(k-1) S_{wst}^2}{N \cdot n} \left[1 + (n-1) \rho_{wst} \right]$$

$$V_{ysy} = \frac{(N-n) S_{wst}^2}{N \cdot n} \left[1 + (n-1) \rho_{wst} \right]$$

This result shows that systematic sample has the same precision as the corresponding stratified random sample with one unit per stratum ^{s_i} if $\rho_{wst} = 0$.

i.e.

$$V_{ysy} = \frac{(N-n) S_{wst}^2}{N \cdot n}$$

Comparison Of Systematic Sampling With Stratified Random Sampling And Simple Random Sampling.

Theorem: 2002 Q7(b) :- + 2004 Q7(b).

If the population follows a linear trend, prove that *

$$V_{st} \leq V_{sy} < V_{sran}$$

Proof: we have

$$V_{sran} = \left(\frac{N-n}{N \cdot n} \right) S^2 \quad \text{where } S^2 = \frac{1}{N-1} \sum (y_i - \bar{y})^2$$

$$V_{sy} = \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \bar{y})^2 \quad S^2 = \frac{1}{N-1} \left[\sum y_i^2 - \frac{(\sum y_i)^2}{N} \right]$$

$$V_{st} = \left(\frac{N-n}{N \cdot n} \right) S_{st}^2$$

we assume y_i as i

$$i = 1, 2, 3, \dots, N$$

$$\sum_{i=1}^N i = 1 + 2 + 3 + \dots + N = \frac{N(N+1)}{2}$$

$$\sum_{i=1}^N i^2 = 1^2 + 2^2 + 3^2 + \dots + N^2 = \frac{N(N+1)(2N+1)}{6}$$

$$S^2 = \frac{1}{N-1} \left[\sum i^2 - \frac{(\sum i)^2}{N} \right]$$

$$S^2 = \frac{1}{N-1} \left[\frac{N(N+1)(2N+1)}{6} - \frac{N^2(N+1)^2}{4N} \right]$$

$$S^2 = \frac{N(N+1)}{N-1} \left[\frac{(2N+1)}{6} - \frac{(N+1)}{4} \right]$$

$$S^2 = \frac{N(N+1)}{N-1} \left[\frac{8N+4-6N-6}{24} \right]$$

$$S^2 = \frac{N(N+1)}{N-1} \left[\frac{2N-2}{24} \right]$$

$$S^2 = \frac{N(N+1)(N-1) \cdot 2}{(N-1) \cdot 12}$$

$$S^2 = \frac{N(N+1)}{12}$$

Thus: Put this value in "Var"

$$Var = \left(\frac{N-n}{N} \right) \frac{N(N+1)}{12}$$

$$Var = \frac{n(k-1)(N+1)}{12n}$$

$$Var = \frac{(k-1)(N+1)}{12} \quad (I)$$

Now;

$$V_{st} = \frac{(N-n)}{N \cdot n} S_{urst}^2$$

To find S_{urst}^2 , we need to replace
N by k in S^2 . Thus

$$S_{urst}^2 = \frac{k(k+1)}{12}$$

$$V_{st} = \frac{(N-n)}{N \cdot n} \cdot \frac{k(k+1)}{12}$$

$$V_{st} = \frac{n(k-1)}{n \cdot nk} \cdot \frac{k(k+1)}{12}$$

$$V_{st} = \frac{(k-1)(k+1)}{12n}$$

Now for systematic sampling the mean of the second sample exceeds that of the first sample by '1'. The mean of the third systematic sample exceeds that of the second sample by '1' and so on.

* Thus the means $\bar{y}_u$ may be replaced by the numbers 1, 2, 3, ..., k. Hence

$$\sum_{u=1}^k (\bar{y}_u - \bar{Y})^2 = \frac{k(k+1)}{2} \cdot \frac{\sum_{i=1}^k i^2}{k} - \left(\frac{\sum_{i=1}^k i}{k} \right)^2$$

I	II	III
1	2	3
4	5	6
7	8	9
4	5	6

mean by 1 and so on. This difference is never be greater than 1.

... $k \rightarrow$ This can be replaced by 1, 2, 3, ..., k

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)(2k+1)}{6} - \frac{k^2(k+1)^2}{4k}$$

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)}{2} \left[\frac{2k+1}{3} - \frac{k(k+1)}{2k} \right]$$

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)}{2} \left[\frac{2k+1}{3} - \frac{k+1}{2} \right]$$

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)}{2} \left[\frac{4k+2-3k-3}{6} \right]$$

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)}{2} \left[\frac{k-1}{6} \right]$$

$$\sum_{u=1}^k (\bar{y}_u - \bar{y})^2 = \frac{k(k+1)(k-1)}{12}$$

Put this value in " V_{sy} "

$$V_{sy} = \frac{1}{k} \sum (\bar{y}_u - \bar{y})^2$$

$$V_{sy} = \frac{1}{k} \cdot \frac{k(k+1)(k-1)}{12}$$

$$V_{sy} = \frac{(k+1)(k-1)}{12} \quad \text{--- (III)}$$

Comparing eq (I) and (III)

$$V_{sy} \leq V_{ran}$$

Now comparing eq. (II) & (III)

$$V_{st} \leq V_{sy}$$

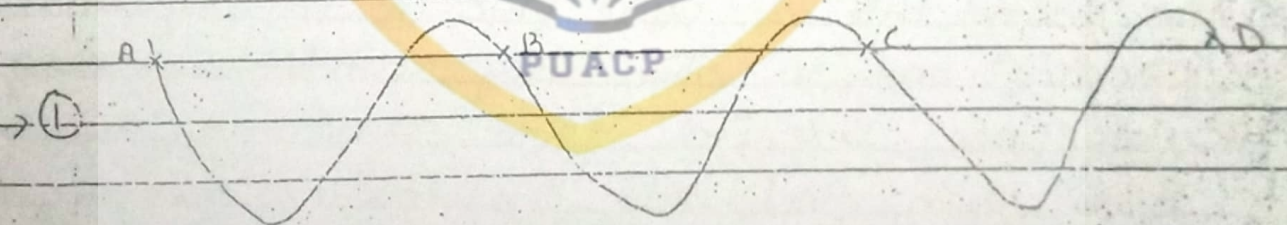
Finally we can write

$$V_{st} \leq V_{sy} \leq V_{ran}$$

Equality occurs if and only if the sample size $n=1$

$$N=nk \quad \because n=1 \\ N=k$$

~~Population with Periodic Variation:~~



$k \rightarrow$ full length of the cycle

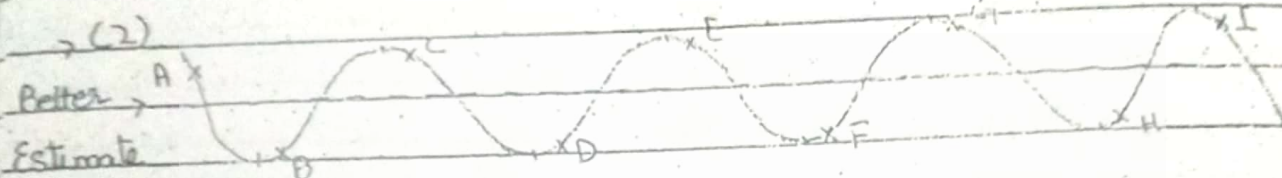
We draw a random number from 1-k

Let suppose the first sampling unit selected is A and that is at peak of cycle then all other sampling units selected are also on the peak of the ^{cycle} samples. But this type of sample will overestimate the mean b/c it consist of values

on peak only and not containing ^{low} sales

Now if we take

$k \rightarrow$ half length of the cycle



We draw a sampling unit from $1-k$. Let the first sampling unit selected is A which is at peak then from next half B is selected which is at low position b/c A is selected from start of the first half cycle than B should also be selected from start of the second half of cycle (b/c it is systematic sampling) and in this way whole sampling units are selected from start of the half of the cycle. And this systematic sample is the better estimate of mean b/c it contains both too high and too low values.

Thus $k = \text{half cycle}$ is better / or must be preferred than $k = \text{full cycle}$

If the population consists of periodic trend e.g. the sine wave, the effectiveness of the systematic sample depends on the value of k (as we discuss in above example).

As in (1)

Sample point A represents the case least favourable to the systematic sample, every observation within the systematic sample is exactly the same, so that the sample is no more precise than a single observation taken at random from the population.

The most favourable case occurs when k is half length

Every systematic sample has a mean exactly equal to the true population mean. Since successive deviations above and below the middle line cancel in (2), the sampling variance of mean is therefore zero.

Also these two cases the sample has various degrees of effectiveness, depending upon the relation of k & λ wavelength.

Population in Random Order:

We define an infinite super population and this superpopulation has a certain properties.

We can draw a finite population of size N . By just changing the codes of the sampling unit, N finite populations can be drawn from the super population.

e.g. $y_1, y_2, y_3, \dots, N$
 $y_2, y_1, y_3, \dots, N$

Following are the properties of super pop.

$$\Rightarrow \sum (y_i) = N\mu$$

$$\Rightarrow \sum (y_i - \mu)^2 = \sigma^2$$

$$\Rightarrow \sum (y_i - \mu)(y_j - \mu) = 0 \text{ for } i \neq j$$

Theorem: 2004 Q7(b)

If the variate y_i ($i=1, 2, \dots, N$) are drawn at random from a super population in which $\sum (y_i) = N\mu$, $\sum (y_i - \mu)^2 = \sigma^2$ and

$$\sum (y_i - \mu)(y_j - \mu) = 0 \text{ for } i \neq j \text{ then}$$

$$\sum (V_{sy}) = \sum (V_{sam})$$

Prove this theorem

Proof: By definition from Anna's Notes

$$V_{sam} = \left(\frac{N-n}{N-n} \right) s^2$$

And $S^2 = \frac{1}{N-1} \sum (y_i - \bar{y})^2$

involve μ or properties of $\sum$
 $\sum (a+b) = \sum a + \sum b$

Consider $\sum (y_i - \bar{y})^2 = \sum (y_i - \mu + \mu - \bar{y})^2$ By adding & subtracting μ

$$= \sum [(y_i - \mu) + (\mu - \bar{y})]^2$$

$$= \sum [(y_i - \mu)^2 + (\mu - \bar{y})^2 + 2(y_i - \mu)(\mu - \bar{y})]$$

$$= \sum (y_i - \mu)^2 + N(\mu - \bar{y})^2 + 2N(\mu - \bar{y})$$

$$= \sum (y_i - \mu)^2 - N(\bar{y} - \mu)^2 \therefore \sum (y_i - \mu)$$

$$\frac{N \sum y_i - N\mu}{N}$$

$$N(\bar{y} - \mu)$$

Consider

$$N(\bar{y} - \mu)^2 = N \left(\frac{\sum y_i}{N} - \mu \right)^2$$

$$= N \left[\frac{1}{N} \sum (y_i - \mu) \right]^2$$

$$= \frac{1}{N} \left[\sum (y_i - \mu)^2 + \sum_{i \neq j} (y_i - \mu)(y_j - \mu) \right]$$

Now:

$$\sum (y_i - \mu)^2 =$$

$$E \left[\sum (y_i - \bar{y})^2 \right] = \frac{1}{N} \left[\sum E (y_i - \mu)^2 + \sum_{i \neq j} E (y_i - \mu)(y_j - \mu) \right]$$

Theorem

using the results stated in the theorem

Notes

$$E \left[\sum (y_i - \bar{y})^2 \right] = \frac{1}{N} \left[\sum \sigma_i^2 + \sum \sigma_j^2 \right]$$

$$E \left[\sum_{i=1}^N (\bar{y}_i - \bar{Y})^2 \right] = \left(\frac{N-1}{N} \right) \sum_{i=1}^N \sigma_i^2$$

Therefore

$$E_p(\text{Var}_n) = \left(\frac{N-n}{N-n} \right) \frac{(N-1)}{N(N-1)} \sum \sigma_i^2$$

$$E_p(\text{Var}_n) = \left(\frac{N-n}{nN^2} \right) \sum \sigma_i^2 \rightarrow (I)$$

Now for Var_y :

$$\text{Var}_y = \frac{1}{K} \sum_{u=1}^K (\bar{y}_u - \bar{Y})^2 \quad (\text{By definition})$$

$$\text{Var}_y = \frac{1}{K} \sum_{u=1}^K \left[(\bar{y}_u - \mu) - (\bar{Y} - \mu) \right]^2 \quad \text{Adding \& Subtracting } \mu$$

$$= \frac{1}{K} \left[\sum_{u=1}^K (\bar{y}_u - \mu)^2 + K(\bar{Y} - \mu)^2 - 2 \sum_{u=1}^K (\bar{y}_u - \mu)(\bar{Y} - \mu) \right]$$

$$= \frac{1}{K} \left[\sum_{u=1}^K (\bar{y}_u - \mu)^2 + K(\bar{Y} - \mu)^2 - 2 \sum_{u=1}^K (\bar{y}_u - \mu)(\bar{Y} - \mu) \right]$$

$$= \frac{1}{K} \left[\sum_{u=1}^K (\bar{y}_u - \mu)^2 - K(\bar{Y} - \mu)^2 \right]$$

Consider;

$$\sum_{u=1}^K (\bar{y}_u - \mu)^2 = \sum_{u=1}^K \left(\frac{\sum_{j=1}^n y_{uj}}{n} - \mu \right)^2$$

(Now we convert ij into i so that we can use given identities)

$$\sum_{u=1}^k (\bar{y}_{u.} - \mu)^2 = \sum_{u=1}^k \left[\frac{1}{n} \sum_j (y_{uj} - \mu) \right]^2 \quad \because i = uj$$

$$= \frac{1}{n^2} \sum_{u=1}^k \left[\sum_j (y_{uj} - \mu)^2 + \sum_{u \neq v} (y_{uj} - \mu)(y_{vj} - \mu) \right]$$

$$= \frac{1}{n^2} \left[\sum_{i=1}^N (y_i - \mu)^2 + \sum_{i \neq j} (y_i - \mu)(y_j - \mu) \right]$$

$$E \left[\sum_{u=1}^k (\bar{y}_{u.} - \mu)^2 \right] = \frac{1}{n^2} \left[\sum_{i=1}^N \sum_j (y_i - \mu)^2 + \sum_{i \neq j} \sum_j (y_i - \mu)(y_j - \mu) \right]$$

$$= \frac{1}{n^2} \sum_{i=1}^N \sigma_i^2$$

For:

$$K(\bar{Y} - \mu) = K \left(\frac{\sum_{i=1}^N y_i}{N} - \mu \right)$$

subtracting

$$= K \left[\frac{1}{N} \sum (y_i - \mu) \right]^2$$

$$K(\bar{Y} - \mu)$$

$$\sum_{u=1}^k (\bar{y}_{u.} - \mu)$$

$$\sum_{u=1}^k \bar{y}_{u.} - k\mu$$

$$K \sum_{u=1}^k \bar{y}_{u.} - k\mu$$

$$= K\bar{Y} - k\mu$$

$$K(\bar{Y} - \mu)$$

$$= \frac{K}{N^2} \left[\sum_{i=1}^N (y_i - \mu)^2 + \sum_{i \neq j} (y_i - \mu)(y_j - \mu) \right]$$

$$\sum [K(\bar{Y} - \mu)] = \frac{K}{N^2} \left[\sum_{i=1}^N \sum_j (y_i - \mu)^2 + \sum_{i \neq j} \sum_j (y_i - \mu)(y_j - \mu) \right]$$

$$= \frac{K}{N^2} \sum_{i=1}^N \sigma_i^2$$

Therefore:

$$\sum (V_{sy}) = \frac{1}{K} \left[\frac{1}{n^2} \sum_{i=1}^N \sigma_i^2 - \frac{K}{N^2} \sum_{i=1}^N \sigma_i^2 \right]$$

$$\Sigma_y (V_{sy}) = \frac{1}{N-n} \sum_{i=1}^N \sigma_i^2 - \frac{1}{N^2} \sum_{i=1}^N \sigma_i^2$$

$$= \left(\frac{N-n}{nN^2} \right) \sum_{i=1}^N \sigma_i^2 \quad \text{--- (II)}$$

Now, comparing the results of (I) & (II), it is proved that

$$\Sigma_y (V_{sy}) = \Sigma_y (V_{sran})$$

That means if the population is in random order then the average simple random sampling and systematic sampling are equally precise.